CampusSphere

Vision & Scope Document

Campus Opportunity Hub

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Revision History

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1. Introduction

1.1 Purpose

The purpose of this document is to collect, analyze, and define the high-level needs and features of CampusSphere, the Campus Opportunity Hub. It focuses on the capabilities needed by the stakeholders and target users, and why these needs exist. The details of how CampusSphere fulfills these needs are detailed in the requirements specification and use-case model.

1.2 Scope

This Vision document applies to CampusSphere, a centralized web-based platform that will be developed as a semester-long Software Engineering project. CampusSphere lets students discover, apply for, and track campus opportunities — internships, hackathons, scholarships, research projects, competitions, workshops, certifications, and club recruitments — through a single system, while faculty, recruiters, and student organizations post and manage those opportunities. This document is associated with the CampusSphere requirements specification (SRS) and the CampusSphere ProjectLibre project plan.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
CampusSphere	The Campus Opportunity Hub platform described by this document.
SRS	Software Requirements Specification.
API	Application Programming Interface.
JWT	JSON Web Token — used for authenticating users.
RBAC	Role-Based Access Control (Student, Faculty, Recruiter, Admin roles).
SDLC	Software Development Life Cycle.
ML	Machine Learning — underlies the recommendation engine.
Opportunity	An internship, hackathon, scholarship, research project, competition, workshop, certification, or club recruitment posted on CampusSphere.

1.4 References

1. CampusSphere Software Requirements Specification (SRS), Project Team.
2. CampusSphere ProjectLibre Document — Work Breakdown Structure, Schedule, and Resource Plan.
3. Vision Document Template (Rational Unified Process).
4. Course Software Engineering Lab assignment specification.

1.5 Overview

The remainder of this Vision document is organized as follows: Section 2 positions CampusSphere in terms of the business opportunity, the problem being solved, and its product position statement. Section 3 describes the stakeholders and users of the system. Section 4 gives a high-level product overview, including product perspective and summary of capabilities. Section 5 lists the product features. Sections 6–8 cover constraints, quality ranges, and feature precedence. Section 9 lists other product requirements, and Section 10 lists documentation requirements.

2. Positioning

2.1 Business Opportunity

Every semester, students miss internships, hackathons, scholarships, and club recruitment deadlines simply because the information is scattered across WhatsApp groups, email threads, LinkedIn posts, departmental notice boards, and word of mouth. Generic job boards and platforms such as LinkedIn

or Handshake are built for the broader job market, not for the mix of academic and extracurricular opportunities specific to a single campus, and they do not understand a student's course load, skill set, or club memberships well enough to recommend the right opportunity at the right time.

CampusSphere addresses this gap by giving a university a single, campus-specific platform where faculty, recruiters, and student organizations post opportunities, and where students discover, bookmark, and apply for them - with a recommendation engine that improves as it learns from an initial skills/interest survey and each student's ongoing activity. Because the project also produces a rich relational database, role-based workflows, and a full SDLC artifact trail, it is well suited both as a semester-long Software Engineering project and as a genuinely useful tool a university could pilot.

2.2 Problem Statement

The problem of	campus opportunities (internships, hackathons, scholarships, research projects, workshops, certifications, and club activities) being scattered across emails, social media, messaging groups, and departmental notices
affects	students across all departments and years, as well as the faculty, recruiters, and student organizations who want to reach them
the impact of which is	students missing deadlines and opportunities that match their skills and interests, and organizers struggling to reach the right audience and track responses
a successful solution would be	a single centralized platform where opportunities can be posted, discovered, filtered, bookmarked, and applied for, with personalized recommendations and deadline reminders that keep students engaged

2.3 Product Position Statement

For	students, faculty, recruiters, and student organizations at a university
Who	need a single place to post, discover, and track campus opportunities
The CampusSphere	CampusSphere is a campus opportunity management platform
That	centralizes internships, hackathons, scholarships, research projects, and club opportunities, and recommends the right ones to each student based on their evolving skill and interest profile
Unlike	generic job boards (LinkedIn, Handshake) or scattered WhatsApp groups, emails, and notice boards
Our product	is built specifically for a single campus community, with role-based posting/approval workflows, application tracking, and a

	recommendation engine seeded by an initial survey and refined through ongoing student activity

3. Stakeholder and User Descriptions

3.1 Market Demographics

CampusSphere targets a single university campus initially, covering all departments and academic years (undergraduate and postgraduate). A mid-size university typically has several thousand students, several hundred faculty members, an active placement/career-services cell, and dozens of recognized student clubs and organizations — all of whom currently rely on fragmented, manual channels to publicize and discover opportunities. The pilot campus's own reputation for placements and student engagement stands to benefit directly from higher visibility and participation in the opportunities it posts.

3.2 Stakeholder Summary

Stakeholder	Description	Key Responsibilities / Interest
University Administration	Owns the institutional relationship and overall platform policy.	Approves the platform's rollout; ensures data-privacy compliance; monitors adoption and impact on student outcomes.
Placement / Career Services Cell	Coordinates recruiter relationships and official placement activity.	Verifies recruiter-posted opportunities; reviews platform analytics; escalates issues to Admin.
Faculty Coordinators	Academic staff who sponsor research projects, competitions, and workshops.	Post and approve opportunities within their department; monitor student applications relevant to their courses.
Student Club Leadership	Runs recognized student organizations and their recruitment drives.	Post club recruitment opportunities; manage applicant pipelines for their club.
Recruiters / Companies	External organizations offering internships or jobs to students.	Post internship/job opportunities; review applications; is billed no fee during the pilot phase.

3.3 User Summary

User	Represents	Key Responsibilities
Student	The primary end user of the platform.	Completes the initial skills/interest survey; searches, filters, bookmarks,

		and applies for opportunities; tracks application status; receives recommendations and deadline reminders.
Faculty	Academic staff posting or approving opportunities.	Posts research/workshop opportunities; reviews and approves postings within their department; reviews applications.
Recruiter	External company representative.	Posts internship/job opportunities; reviews applicant profiles and resumes; updates application status.
Admin	Platform/University administrator.	Approves organizations and opportunity postings; manages categories and tags; monitors platform-wide analytics and moderates content.

3.4 User Environment

- Students typically check the platform from a mobile browser between classes and from a laptop during focused application sessions, so the UI must work well on both.
- An opportunity's lifecycle — posting, discovery, application, and decision — usually plays out over days to a few weeks, matching application deadlines.
- Peak load occurs at the start of each semester (recruitment/hackathon season) and just before major scholarship or internship deadlines.
- Students already use email, WhatsApp, Instagram, and LinkedIn; CampusSphere should complement these (e.g. via email/SMS notifications) rather than require abandoning them.
- Faculty and recruiters are occasional users who post and review periodically rather than daily, so the posting/approval workflow must be quick and low-friction.

3.5 Stakeholder Profiles

3.5.1 Placement / Career Services Cell

Description	The university office responsible for coordinating student placements and recruiter relationships.
Type	Business / semi-technical; comfortable with spreadsheets and email but not a developer.
Responsibilities	Verify recruiter accounts and postings; monitor placement-related analytics; act as the escalation point for recruiter issues.
Success Criteria	Higher visibility and turnout for recruiter-posted opportunities; fewer missed deadlines reported by students.
Involvement	Requirements reviewer for recruiter-facing and analytics features; approves the recruiter role's permissions.
Deliverables	Periodic placement-activity summary exported from the platform's analytics dashboard.

3.6 User Profiles

3.6.1 Student

Description	An undergraduate or postgraduate student at the university; the platform's primary end user.
Type	Casual user; expects a simple, mobile-friendly experience similar to consumer apps.
Responsibilities	Completes the initial survey and skill profile; searches, filters, bookmarks, and applies for opportunities; keeps their resume and profile current.
Success Criteria	Discovers relevant opportunities faster than before; does not miss deadlines; receives recommendations that feel genuinely relevant.
Involvement	Primary source of usability feedback during UAT.
Deliverables	None — consumes the system rather than producing artifacts for it.

3.7 Key Stakeholder or User Needs

Need	Reason	How Solved Today	What They Want

Discover relevant opportunities quickly	Opportunities are scattered across many channels	Manually checking WhatsApp groups, email, and notice boards	One searchable, filterable list with personalized recommendations
Avoid missing deadlines	No centralized reminder system exists today	Word of mouth or accidental discovery	Deadline reminders and notifications
Reach the right students	Faculty/recruiters/clubs struggle to target the right audience	Mass emails or generic social posts with low targeting	Posting with category/tag targeting and applicant analytics
Track application status	No visibility into where an application stands	Emailing the organizer directly to ask	An in-app application-tracking dashboard

3.8 Alternatives and Competition

3.8.1 WhatsApp / Email / Notice Boards

The current status quo. Zero cost and universally adopted, but unsearchable, easy to miss, and impossible to personalize or track applications through.

3.8.2 LinkedIn / Handshake / Unstop

General-purpose or multi-campus platforms with far larger opportunity catalogs, but they are not tailored to this specific campus's clubs, courses, or faculty-run opportunities, and they don't integrate with the university's own approval workflows.

4. Product Overview

4.1 Product Perspective

CampusSphere is a new, independent web-based platform; it is not a component of an existing university system, although it is expected to eventually integrate with the university's single sign-on and email systems. Students, faculty, and recruiters interact with the platform through a web/mobile client, while university administrators use a dedicated admin console. Both clients communicate with the CampusSphere platform, which comprises the API layer, the PostgreSQL database, and the recommendation engine.

4.2 Summary of Capabilities

Customer Benefit	**Supporting Features**
Students discover relevant opportunities faster.	Search and filtering, skill-based tagging, categories, and an AI-powered recommendation engine seeded by an initial survey.
Students never miss a deadline.	Bookmarks/favorites, deadline reminders, and a notification system.
Students can track where each application stands.	Application tracking dashboard with status updates.
Faculty, recruiters, and clubs reach the right students.	Opportunity posting with category/tag targeting and an approval workflow.
The recommendation engine keeps improving.	Continuous feedback loop that refines each student's profile based on applications, bookmarks, and survey responses.
Administrators can monitor platform health and engagement.	Analytics dashboard covering postings, applications, and user activity.

4.3 Assumptions and Dependencies

1. It is assumed that students, faculty, and recruiters have reliable access to a smartphone or computer with a modern web browser and internet access.
2. It is assumed that the university will designate an Admin user (or team) to approve organizations and moderate postings.

3. It is assumed that recruiters and faculty will complete a lightweight verification step before their postings go live, to reduce the risk of spam or fraudulent listings.

4. The default language for CampusSphere is English; multilingual support is out of scope for the initial release.

5. The quality of recommendations depends on students completing the initial survey and continuing to interact with the platform (applying, bookmarking); users who do not will receive only generic, popularity-based recommendations.

6. It is assumed that the university can provide (or the project can provision) cloud hosting for the pilot deployment.

4.4 Cost and Pricing

CampusSphere is developed as an academic Software Engineering project and, during the pilot phase, will be offered to the university free of charge. Ongoing costs are limited to cloud hosting (web/database servers, backups) and, if enabled, third-party AI API usage for recommendation generation; both are estimated in the accompanying ProjectLibre resource and cost plan. No student, faculty, or recruiter will be charged to use the platform.

4.5 Licensing and Installation

CampusSphere is delivered as a hosted web application (Software-as-a-Service); there is no client-side installation beyond a modern web browser. The backend is containerized with Docker and deployed to a cloud provider (e.g. Render, Railway, or AWS), so installation is limited to one-time environment provisioning and configuration by the DevOps engineer, not to end-user setup.

5. Product Features

5.1 Account & Role Management

1. Register and log in securely (JWT-based authentication)
2. Role-based access for Student, Faculty, Recruiter, and Admin
3. Create and edit skill profile
4. Upload and update resume
5. Complete the initial onboarding survey (interests, skills, goals)

5.2 Opportunity Discovery

1. Search opportunities by keyword
2. Filter opportunities by category, tag, skill, and deadline
3. View opportunity details (organization, description, deadline, eligibility)
4. Bookmark / favorite an opportunity
5. Receive AI-powered personalized recommendations

5.3 Application & Tracking

1. Apply to an opportunity
2. View application status (applied, under review, shortlisted, rejected, selected)
3. View a dashboard of all past and current applications
4. Comment on / ask questions about an opportunity

5.4 Opportunity Posting & Approval

1. Post a new opportunity (Faculty, Recruiter, or Club Admin)
2. Submit a posting for Admin approval
3. Approve, reject, or request changes to a posting (Admin)
4. Categorize and tag an opportunity
5. Edit or close an opportunity

5.5 Recommendation Engine

1. Generate an initial recommendation set from the onboarding survey
2. Continuously refine recommendations based on applications, bookmarks, and profile updates

3. Rank recommended opportunities by predicted relevance

5.6 Notifications & Reminders

1. Send deadline-reminder notifications
2. Send status-change notifications (application update, posting approved/rejected)
3. View an in-app notification center

5.7 Admin & Analytics

1. Approve or suspend organizations and users
2. Manage categories and tags
3. View platform-wide analytics (postings, applications, active users)
4. Moderate comments and reported content

6. Constraints

- The system must be delivered within the academic semester's project timeline (see the CampusSphere ProjectLibre schedule).
- The team has a fixed set of student developers and no dedicated budget for paid third-party services beyond free tiers.
- The initial release targets English-language content only.
- The recommendation engine's quality is bounded by the amount of survey and interaction data available, especially in the first weeks of the pilot (the “cold-start” problem).
- The backend must be built with FastAPI and PostgreSQL, and the frontend with React.js, per the team's chosen technology stack.

7. Quality Ranges

- Usability: a new student should be able to complete the onboarding survey and submit their first application within 10 minutes, without external help.
- Performance: search and filter results should return within 2 seconds under normal pilot load (a few hundred concurrent users).
- Availability: the platform should target 99% uptime during active recruitment periods (start of semester, major deadlines).
- Reliability: no application or bookmark data should be lost once submitted; all writes are persisted to the primary database with regular backups.
- Security: passwords are never stored in plaintext; all authenticated endpoints require a valid JWT; role permissions are enforced server-side, not just in the UI.

8. Precedence and Priority

Priority	Features
Critical	Authentication & roles; opportunity posting & approval; search & filtering; application submission & tracking; core database schema.
Important	Bookmarks, notifications & deadline reminders, skill profile, resume upload, admin analytics dashboard.
Useful	AI-powered recommendation engine, continuous feedback loop, comments on opportunities, achievements/badges.

Critical features are essential — the pilot cannot launch without them. Important features significantly affect student and organizer satisfaction but would not block a launch. Useful features, including the evolving recommendation engine, add differentiating value and can be delivered incrementally after the core platform is stable.

9. Other Product Requirements

9.1 Applicable Standards

CampusSphere should follow standard web accessibility guidance (WCAG 2.1 AA where practical) and use HTTPS/TLS for all traffic. Password storage must follow standard hashing practice (e.g. bcrypt), and the system should be designed with the university's data-privacy obligations for student data in mind.

9.2 System Requirements

The web/mobile client requires only a modern evergreen browser (Chrome, Edge, Firefox, Safari) with JavaScript enabled; no native app installation is required. The backend requires a Linux host capable of running Docker containers for the FastAPI service and the PostgreSQL database, as detailed in the CampusSphere ProjectLibre resource plan.

9.3 Performance Requirements

The platform should support at least a few hundred concurrent users during peak recruitment periods without noticeable degradation, with search/filter and dashboard page loads returning within 2 seconds under normal conditions.

9.4 Environmental Requirements

As a cloud-hosted web application, CampusSphere has no special hardware environmental requirements beyond standard data-center conditions for its cloud provider. On the client side, the responsive UI should remain usable on both desktop and mobile screen sizes and under typical campus Wi-Fi conditions.

10. Documentation Requirements

10.1 User Manual

A short online user guide will cover account creation, completing the onboarding survey, searching and applying for opportunities, and tracking application status — written for a student audience with no technical background.

10.2 Online Help

Contextual tooltips and a lightweight in-app FAQ/help section will cover common tasks for each role (Student, Faculty, Recruiter, Admin), reducing the need for a separate help desk during the pilot.

10.3 Installation Guides, Configuration, and Read Me File

A README and deployment guide will document environment setup, environment variables, database migration steps, and Docker-based deployment, so the DevOps engineer (or a future maintainer) can reproduce the production environment.

10.4 Labeling and Packaging

The platform will use a consistent CampusSphere logo, color scheme, and naming across the web client, admin console, and any notification emails, to give the product a recognizable, professional identity on campus.

Appendix A: Feature Attributes

Features are given attributes that can be used to evaluate, track, prioritize, and manage the product items proposed for implementation. The following attributes are used to manage the CampusSphere feature set:

A.1 Status

Set after negotiation and review by the project management team, this attribute tracks progress during the definition of the project baseline:

- Proposed: Used to describe features that are under discussion (such as achievements or badges) but have not yet been reviewed and accepted by the project team or university stakeholders.
- Approved: Capabilities that are deemed useful and feasible, and have been approved for implementation by the official channel.
- Incorporated: Features incorporated into the product baseline at a specific point in time.

A.2 Benefit

Set by the project team or business analyst to rank requirements by their relative benefit to the end users. For CampusSphere, this corresponds to the Critical, Important, and Useful priorities outlined in Section 8.

A.3 Effort

An estimation of the time, team bandwidth, or resources required to implement a specific feature within the fixed academic semester's project timeline.

A.4 Risk

The probability of technical complexities, dependencies, or scheduling challenges affecting the successful delivery of a feature. For example, the "cold-start" problem associated with the recommendation engine carries a higher risk.

A.5 Stability

An assessment of how likely the feature's core requirements or underlying technologies (such as FastAPI, PostgreSQL, or React.js) are to change during the development lifecycle.

A.6 Target Release

Identifies the specific release version or milestone (e.g., the initial pilot phase) in which the feature will be deployed to the university campus.

A.7 Assigned To

The specific project team member (Rajwardhan Singh, Ramitha Ramesh, or Rayi Charan Harsha) responsible for developing, testing, or managing the feature.

A.8 Reason

The underlying justification for including the feature, tying it directly back to solving stakeholder needs, such as helping students track applications or avoiding missed deadlines.